

Multifractal Voice Analysis for Detection of Vocal Pathologies

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Abstract—The classification of voice disorders is problematic due to the complexity of the structure of the voice production mechanism as well as associated with the neurophysiological system changes in this case, which complicates the implementation of an objective classification algorithm. Conventional acoustic measures have proved themselves good in the description of such properties as the basic tone quality of voice, but do not adequately describe the nonlinearity and multi-scale character inherent in pathological speech sounds. The study is devoted to the application of fractal and multifractal analysis to modeling the complexity of voice signal structure. The database contains samples of sustained vowels /a/ of both healthy individuals and people suffering from various kinds of structural and neurological voice disorders. Basic acoustic measures, fractal dimension estimates, multiresolution fractal descriptors using wavelets, and multifractal parameter sets obtained via Multifractal Detrended Fluctuation Analysis are used in this study.

Index Terms—Voice Pathology Detection, Multifractal Analysis, Fractal Dimension, MF-DFA, Saarbrücken Voice Database, Nonlinear Speech Analysis

I. INTRODUCTION

Voice disorders are prevalent among the population and can occur because of structural problems in the vocal folds, neurological problems that affect muscle coordination, and various functional and psychogenic factors. The proper diagnosis usually requires a laryngoscopy procedure together with a perceptual assessment of voice disorders conducted by physicians. Even though these techniques are quite accurate, they can be subjective, time-consuming, and necessitate expert knowledge in medicine.

Recently, there has been a considerable interest in the application of computational algorithms for voice disorder detection and screening as an alternative to traditional clinical diagnosis. Currently available computational techniques for identifying pathological speech mostly employ conventional acoustic features such as MFCCs, fundamental frequency (F_0), and other spectral-based descriptors. Such techniques have

proven to be rather successful for capturing the spectral properties of speech but fail to adequately describe the temporal and nonlinear dynamics associated with pathological voices.

Speech production can be considered as a nonlinear phenomenon governed by a wide range of processes involving airflow, vocal fold vibration, and vocal tract movements. In some cases, these processes may become irregular leading to changes in the dynamics. For a comprehensive analysis of such dynamics, the fractal and multifractal theory can be used as they offer a possibility of studying complex signals in terms of their scaling characteristics.

In this paper, we propose the use of **multifractal detrended fluctuation analysis** technique for assessing speech recordings in the presence of pathologies. Multifractal analysis of time series represents a mathematical methodology aimed at quantifying heterogeneity of dynamics using the concept of scaling behaviour at multiple time scales. Unlike monofractals which quantify signals by means of one scaling exponent, multifractality offers more flexible representation of signal complexity by providing a multifractal spectrum which includes local scaling exponents for various q -orders.

The main purpose of the study is to test whether multifractal descriptors obtained via MF DFA can help detect pathological speech, and whether such descriptors can complement conventional acoustic features.

II. RELATED WORK

Automatic detection of voice pathologies has traditionally relied on acoustic features such as MFCCs, perturbation measures, and spectral statistics combined with statistical or machine learning classifiers [5], [6]. These descriptors capture important spectral and phonatory characteristics, but they are less suited for representing the nonlinear and multi-scale temporal irregularities often present in pathological voices. Nonlinear analysis methods have therefore also been explored

for pathological speech, with prior work showing that dynamical and fractal descriptors can reveal information not captured by conventional acoustic features [4], [8].

Multifractal detrended fluctuation analysis (MFDFA) extends this direction by modelling scaling behaviour across multiple moments and time scales [2]. Multifractal methods have been effective in other biomedical domains for characterising irregular physiological signals [7], but their use in voice pathology detection remains relatively underexplored. This motivates our investigation of whether multifractal descriptors can improve pathological voice detection when combined with standard acoustic and paralinguistic features under speaker-aware evaluation.

III. DATASET

The experiments in this study use recordings from the **Saarbrücken Voice Database (SVD)**, a widely used clinical corpus containing healthy and pathological voices. The original audio is stored in proprietary `.nsp` format and is converted to standard `.wav` for preprocessing and feature extraction. Although SVD contains multiple vowels and connected-speech recordings, this work focuses on the sustained vowel /a/ at normal pitch (`a_n`) because it provides a stable phonation condition for speaker-aware comparison.

Table I summarises the dataset used in the experiments.

TABLE I
DATASET STATISTICS

Statistic	Value
Total recordings (all tokens)	23,119
Unique speakers	1,355
Pathology classes	12 (+ healthy)
Samples for token (<code>a_n</code>)	1,656
Sex distribution	65% female, 35% male

The `a_n` subset is class-imbalanced, which is typical for clinical speech datasets. Very small classes are excluded from grouped experiments, and individual diseases are merged into broader clinical categories when evaluating disease-group classification.

IV. FEATURE EXTRACTION

To capture both conventional acoustic properties and nonlinear signal dynamics, multiple feature families are extracted from each audio sample. In total, approximately **191 features** are computed for every recording. These features include traditional spectral descriptors, multifractal characteristics, standardised paralinguistic features, and nonlinear signal complexity measures.

A. Signal Representation

Before feature extraction, the speech recordings are analysed in both the time and time–frequency domains to capture temporal stability, harmonic structure, and spectral energy distribution across healthy and pathological voices.

Waveform representations reveal variations in amplitude modulation and phonatory stability, while spectrograms highlight differences in harmonic structure and spectral energy distribution. These visual patterns motivate the extraction of both spectral and nonlinear complexity-based features.

B. Acoustic Features

Conventional acoustic features are extracted using the *Librosa* library. The feature set contains **84 descriptors**, including time-domain statistics, spectral moments, zero-crossing measures, MFCC summaries, and fundamental frequency estimates. These features serve as the baseline representation of spectral and phonatory structure.

C. Multifractal Features

To capture nonlinear and multiscale dynamics present in pathological speech signals, multifractal descriptors are extracted using **Multifractal Detrended Fluctuation Analysis (MFDFA)**. From the multifractal spectrum, **12 statistical descriptors** are derived from the generalised Hurst exponent, scaling exponent, and singularity spectrum. The final MFDFA configuration uses first-order detrending, a q range of $[-5, 5]$ with step 0.5, and 40 analysis scales.

D. OpenSMILE Features

In addition to the Librosa-based descriptors, features from the **extended Geneva Minimalistic Acoustic Parameter Set (eGeMAPSv02)** are extracted using the OpenSMILE toolkit. This standardised feature set contributes **88 parameters** capturing prosodic, spectral-shape, voice-quality, and temporal characteristics relevant to clinical speech analysis.

E. NeuroKit2 Features

To further characterise nonlinear signal dynamics and complexity, **seven additional features** are extracted using the NeuroKit2 library, including entropy, fractal-dimension, skewness, and kurtosis measures. These features complement the multifractal descriptors by capturing additional aspects of signal irregularity and statistical structure.

V. DATA PIPELINE AND SPLITTING STRATEGY

This section summarises the preprocessing pipeline used to construct the final machine learning dataset and the safeguards adopted to ensure leakage-free evaluation. The overall workflow is illustrated in Figure 1. Particular attention is given to preventing speaker-level data leakage, which is a common issue in voice pathology datasets. The raw `.nsp` recordings are converted to `.wav`, indexed through a metadata manifest, normalised, and then processed by the four feature-extraction modules before the resulting feature tables are merged into a unified machine learning dataset. To minimise variability and within-speaker duplication, most experiments use only the sustained vowel `a_n`.

A critical concern in clinical voice datasets is **speaker-level leakage**. We therefore use `StratifiedGroupKFold` with speaker identity as the grouping variable so that all recordings from a speaker remain in the same fold. Rare

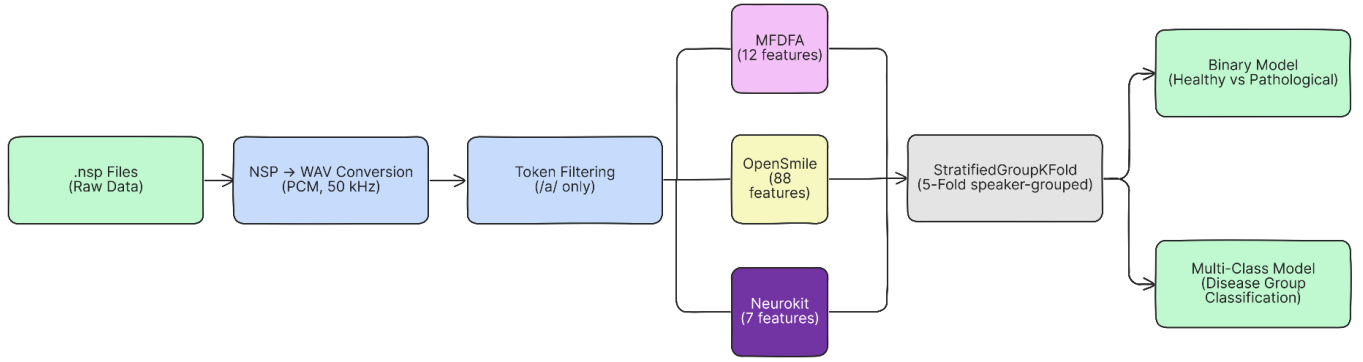


Fig. 1. Overview of the preprocessing and feature extraction pipeline. Raw NSP recordings are converted to WAV format, metadata manifests are generated, and sustained vowel tokens are selected. Multiple feature families are then extracted and merged before training classification models using speaker-grouped cross-validation.

classes are removed when necessary for stable training, and class imbalance is handled through filtering or downsampling of dominant groups. Final performance is reported as the average over five speaker-grouped folds.

VI. EXPERIMENTAL SETUP AND RESULTS

This section presents a comprehensive evaluation of the proposed voice pathology detection framework. The experiments are designed to assess the discriminative capability of multifractal features and to examine their contribution when combined with established acoustic descriptors.

Particular attention is given to ensuring reliable evaluation by avoiding speaker-level data leakage, which is a common issue in voice pathology datasets. All experiments therefore employ a speaker-aware evaluation strategy using five-fold cross-validation with `StratifiedGroupKFold`. This ensures that all recordings from a given speaker appear exclusively in either the training or test split, but never both.

Performance is evaluated using three metrics: accuracy, balanced accuracy, and macro-averaged F1 score. Balanced accuracy and macro F1 are particularly important for this task due to class imbalance and the clinical importance of detecting pathological voices.

Naive random splits produced unrealistically high scores in early experiments, confirming that speaker leakage can strongly inflate apparent performance. All results reported below therefore use the speaker-aware evaluation protocol.

A. Speaker-Aware Evaluation

To obtain more reliable performance estimates, all subsequent experiments employ speaker-aware cross-validation. Only the sustained vowel token `a_n` is used to ensure consistency across speakers and to minimise variability introduced by different speech tasks.

Two classification tasks are considered:

- Binary classification: distinguishing healthy speakers from pathological speakers.
- Disease-group classification: distinguishing neurological disorders from structural disorders.

The binary classification task serves as a screening scenario where the goal is to detect the presence of a voice disorder. Table II summarises the results obtained using five-fold speaker-grouped cross-validation.

TABLE II
BINARY CLASSIFICATION RESULTS (SPEAKER-GROUPED CV)

Model	Accuracy	Balanced Acc.	F1 Macro
LightGBM	0.889	0.875	0.882
XGBoost	0.886	0.868	0.878
Random Forest	0.885	0.865	0.875
Logistic Regression	0.875	0.867	0.870

Compared with the baseline experiment, the performance values decrease substantially, reflecting the more realistic evaluation setting. Gradient boosting models (LightGBM and XGBoost) achieve the highest performance, indicating that nonlinear feature interactions play an important role in distinguishing pathological speech.

The confusion matrices aggregated across the cross-validation folds are shown in Figure 2. These results indicate that the models achieve high recall for pathological voices while maintaining strong precision for the healthy class.

B. Disease Group Classification

Beyond simple pathology detection, a more challenging task involves distinguishing between different types of voice disorders based on their clinical origin. In this work, individual pathologies are grouped into two clinically meaningful categories: neurological disorders and structural disorders.

Neurological disorders typically arise from impairments in neural control of the vocal folds, whereas structural disorders involve physical abnormalities in the vocal fold tissue. Distinguishing between these categories from speech signals alone is therefore a more difficult problem.

Table III summarises the performance of the evaluated models for disease-group classification.

The results indicate that disease-group classification is substantially more difficult than binary pathology detection.

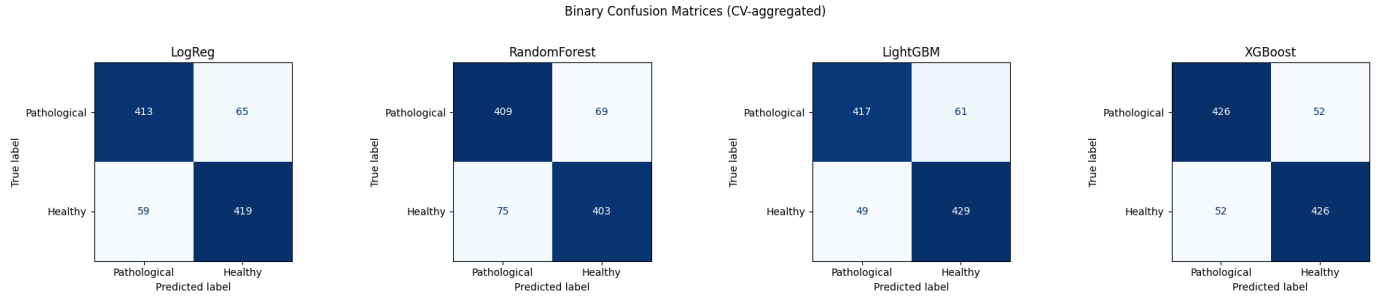


Fig. 2. Cross-validation aggregated confusion matrices for binary classification across different models.

TABLE III
DISEASE GROUP CLASSIFICATION RESULTS

Model	Accuracy	Balanced Acc.	F1 Macro
Random Forest	0.646	0.628	0.629
LightGBM	0.642	0.639	0.637
XGBoost	0.611	0.604	0.603

Even the best-performing model achieves an F1 score of approximately 0.63, reflecting the subtler acoustic differences between neurological and structural disorders.

Figure 3 illustrates the confusion matrices for this task.

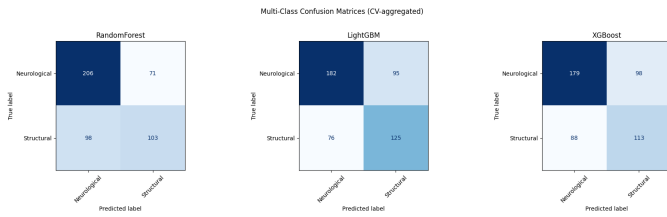


Fig. 3. Confusion matrices for disease-group classification.

C. Feature Ablation Analysis

To better understand the contribution of different feature families, a feature ablation study was conducted. Various combinations of multifractal features, OpenSMILE descriptors, and NeuroKit2 features were evaluated using the same speaker-grouped cross-validation protocol.

TABLE IV
FEATURE ABLATION RESULTS (BINARY CLASSIFICATION)

Feature Configuration	Best Model	Bal. Acc.	F1 Macro
MF DFA + OpenSMILE	LightGBM	0.879	0.878
MF DFA + NeuroKit2 + OpenSMILE	XGBoost	0.876	0.876
MF DFA + NeuroKit2	LogReg	0.868	0.867
MF DFA only	LogReg	0.867	0.866
MF DFA + OpenSMILE — Age	LightGBM	0.752	0.749

The results reveal several interesting observations. First, multifractal features alone achieve an F1 score of 0.866,

demonstrating that multifractal characteristics of the speech signal carry substantial information about vocal health. When combined with OpenSMILE descriptors, performance improves slightly, indicating that these features provide complementary information.

In contrast, adding NeuroKit2 features yields only marginal improvement, suggesting that most of the useful complexity-related information is already captured by the multifractal descriptors.

An additional observation concerns the role of age metadata. When age is removed from the MF DFA+OpenSMILE configuration, macro F1 drops from 0.878 to 0.749, indicating that age is a strongly correlated variable in this dataset and should be interpreted carefully when assessing pathology-detection performance.

The comparative results across configurations are illustrated in Figure 4.

D. Per-Disease Binary Detection

Finally, additional experiments were conducted to evaluate the detectability of individual pathologies. In this setting, a separate binary classifier was trained for each disease against the healthy class.

TABLE V
PER-DISEASE BINARY DETECTION RESULTS

Disease	Best Model	Bal. Acc.	F1 Macro
Recurrent Laryngeal Nerve Paralysis	XGBoost	0.866	0.865
Reinke's Edema	LightGBM	0.829	0.828
Laryngitis	XGBoost	0.796	0.796
Vocal Fold Polyp	LightGBM	0.795	0.793
Functional Dysphonia	XGBoost	0.794	0.794
Psychogenic Dysphonia	XGBoost	0.790	0.788
Hyperfunctional Dysphonia	XGBoost	0.773	0.773
Spasmodic Dysphonia	LightGBM	0.617	0.587
Vocal Fold Nodules	Random Forest	0.531	0.485

The results show that certain disorders, such as recurrent laryngeal nerve paralysis and Reinke's edema, are relatively easier to detect from acoustic signals. In contrast, conditions such as vocal fold nodules and spasmodic dysphonia remain challenging due to their subtle acoustic manifestations and limited sample sizes.

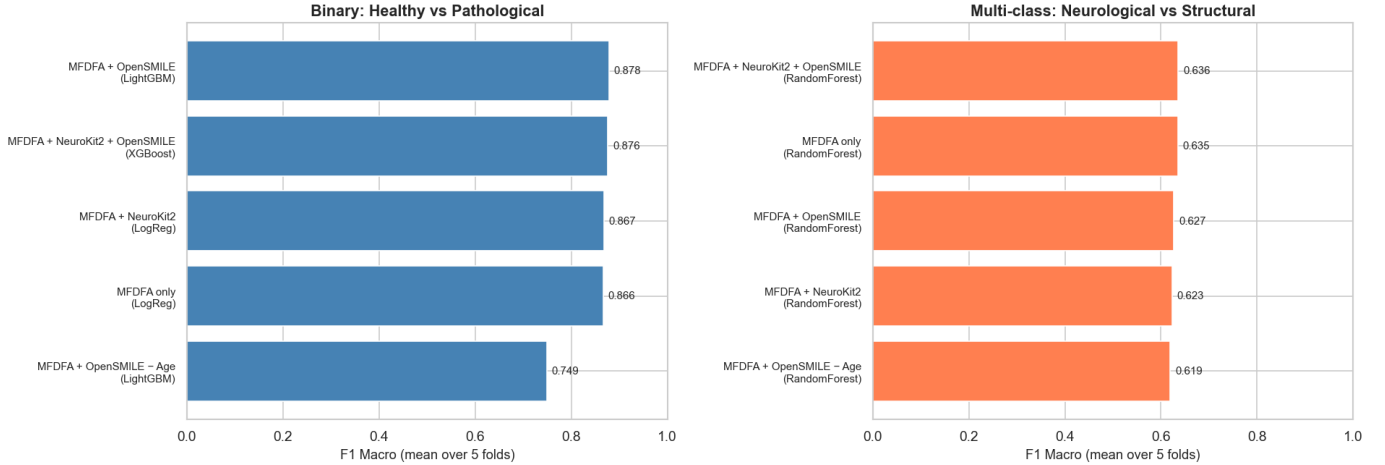


Fig. 4. Feature ablation results for binary and multi-class classification.

These findings suggest that while automated detection of general voice pathology is feasible, reliable classification of specific disorders remains an open research problem requiring larger datasets and more specialised features. More broadly, the experiments highlight three key observations: speaker-aware validation is essential to avoid overly optimistic results, multifractal descriptors alone already provide strong discriminative power, and higher-level pathology screening is currently more reliable than fine-grained disease identification.

VII. CONCLUSION

This paper explores the use of multifractal analysis for the automated detection of voice pathology based on sustained vowel recordings from the Saarbrücken Voice Database. The key focus of this study was the potential use of multifractal descriptors obtained through Multifractal Detrended Fluctuation Analysis (MFDFA) to characterise clinically important features of pathological voices and improve performance of the classifier.

To integrate multifractal descriptors within the feature space, this work proposes combining them with traditional acoustic features (obtained using OpenSMILE) and complexity-based features (obtained using NeuroKit2). The major attention was paid to preventing data leakage during evaluation by utilising speaker-grouped cross-validation and thereby obtaining a valid assessment of model generalisability to novel speakers.

Based on the experimental evaluation performed, the proposed approach demonstrates a good capacity for binary classification of normal and pathological voices yielding a macro F1 score of about **0.89**. Most importantly, according to the experiments on the ablation of individual features, multifractal descriptors are quite effective in their own right as predictors of the presence of voice pathology, indicating valuable temporal information contained in them.

Further experiments show that while binary pathology detection is rather manageable, fine-grained disease classification appears to be more difficult due to acoustic overlap between diseases. In conclusion, this study indicates that multifractal descriptors provide a promising complement to traditional spectral features for automated voice-pathology screening. Future work should validate the approach on larger independent cohorts, explore multi-token and end-to-end models, and study which multifractal descriptors are most clinically informative.

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